

Changhun Kim

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Research Interests

My research focuses on the algorithmic and theoretical foundations of **deep generative models**, particularly **diffusion models**: why they **generalize**, when they **hallucinate**, and how to make them more **trustworthy**. I am also interested in **embodied AI**, where diffusion models drive **world modeling** and **action generation**.

Keywords: Deep Generative Models, Embodied AI, Trustworthy Machine Learning, Machine Learning Theory

Education

University of Wisconsin–Madison Madison, WI, USA
Ph.D. in Computer Science (Advisor: Prof. Grigorios Chrysos) Aug. 2026 – Present

Korea Advanced Institute of Science and Technology (KAIST) Daejeon, Korea
M.S. in Artificial Intelligence (Advisor: Prof. Eunho Yang) Feb. 2022 – Feb. 2024
B.S. in Computer Science and Mathematics Feb. 2017 – Feb. 2022

Employment

UW–Madison Chrysos Lab Madison, WI, USA
Graduate Research Assistant (Advisor: Prof. Grigorios Chrysos) Aug. 2026 – Present

Google Mountain View, CA, USA
Research Intern Jun. 2027 – Sept. 2027

UBC Computer Vision Lab Vancouver, BC, Canada (Remote)
External Collaborator (Advisor: Prof. Evan Shelhamer) Jul. 2025 – Present

AITRICS Seoul, Korea
Researcher (Advisor: Prof. Eunho Yang, Prof. Juho Lee) Nov. 2023 – Jun. 2026

KAIST Machine Learning and Intelligence Lab Daejeon, Korea
Graduate Research Assistant (Advisor: Prof. Eunho Yang) Feb. 2022 – Feb. 2024
Undergraduate Research Assistant (Advisor: Prof. Eunho Yang) Jun. 2021 – Feb. 2022

KAIST Vehicular Intelligence Lab Daejeon, Korea
Undergraduate Research Assistant (Advisor: Prof. Dongsoo Har) Oct. 2019 – Aug. 2020

Publications

Manuscripts *: Equal Contribution, †: Equal Advising

[7] **Some Topics Related To Trustworthy Diffusion World Action Models**

Changhun Kim, Grigorios Chrysos

Under Review at Conference on Neural Information Processing Systems (NeurIPS), 2027

[6] **A Unified Framework for Understanding Generalization and Hallucination in Diffusion Models**

Changhun Kim, Abhinav Narayan Harish, Grigorios Chrysos

Under Review at International Conference on Machine Learning (ICML), 2027

- [5] **EvergreenTTA: Evergreen Test-Time Adaptation under Non-Stationary Long-Horizon Streams**
Changhun Kim, Grigorios Chrysos, Evan Shelhamer
Under Review at IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), 2027
- [4] **ChronoFlow: Hierarchical Flow Matching for Irregular Time Series Generation**
Changhun Kim*, Sunguk Jang*, Jeongjun Lee, Juhwan Choi, Sangchul Hahn, Grigorios Chrysos, Eunho Yang[†], Juho Lee[†]
Under Review at International Conference on Learning Representations (ICLR), 2027
- [3] **Domain Generalization under Sampling Pattern Shifts in Irregularly Sampled Time Series**
Changhun Kim, Joohyung Lee, Kwanhyung Lee, Donghwee Yoon, Grigorios Chrysos, Eunho Yang
Under Review at International Conference on Learning Representations (ICLR), 2027
- [2] **TRIAGE: Dialectical LLM Reasoning for Explainable Risk Prediction on Irregularly Sampled Medical Time Series**
Hyeongwon Jang*, Gyouk Chu*, Changhun Kim, Hangyul Yoon, Jeonguk Lee, Eunho Yang, Joonhyung Park
Under Review at International Conference on Learning Representations (ICLR), 2027
- [1] **Development and External Validation of a Transformer-Based Cardiac Arrest Prediction System for General Ward Patients: A Multicenter Study**
Sungsoo Hong*, Kyung Hyun Lee*, Yechan Mun*, Sangchul Hahn, Sunguk Jang, Changhun Kim, Yuhyun Choi, Kyung Soo Chung, Kang-Mo Gu, Hae In Jung, Tae Wan Kim, Moon Seong Baek, Taeyong Sim
Under Review at BMC Medical Informatics and Decision Making, 2026

Conference Publications

- [8] **ReviewScore: Misinformed Peer Review Detection with Large Language Models**
Hyun Ryu, Doohyuk Jang, Hyemin S. Lee, Joonhyun Jeong, Gyeongman Kim, Donghyeon Cho, Gyouk Chu, Minyeong Hwang, Hyeongwon Jang, Changhun Kim, Haechan Kim, Joowon Kim, Yoonjeon Kim, Kwanhyung Lee, Chanjae Park, Heecheol Yun, Gregor Betz, Eunho Yang
Conference on Empirical Methods in Natural Language Processing (EMNLP), 2026 (Findings)
- [7] **Delta-XAI: A Unified Framework for Explaining Prediction Changes in Online Time Series Monitoring**
Changhun Kim*, Yechan Mun*, Hyeongwon Jang, Eunseo Lee, Sangchul Hahn[†], Eunho Yang[†]
International Conference on Learning Representations (ICLR), 2026
- [6] **Soft Equivariance Regularization for Invariant Self-Supervised Learning**
Joohyung Lee, Changhun Kim, Hyunsu Kim, Kwanhyung Lee, Juho Lee
International Conference on Learning Representations (ICLR), 2026
- [5] **Position Paper: How Should We Responsibly Adopt LLMs in the Peer Review Process?**
Juhwan Choi, JungMin Yun, Changhun Kim, YoungBin Kim
Conference of the European Chapter of the Association for Computational Linguistics (EACL), 2026 (Findings)
- [4] **TIMING: Temporality-Aware Integrated Gradients for Time Series Explanation**
Hyeongwon Jang*, Changhun Kim*, Eunho Yang
International Conference on Machine Learning (ICML), 2025 (Spotlight, 313/12,107=2.6%)
ICLR Workshop on XAI4Science, 2025
CIKM Workshop on Human-Centric AI, 2025 (Best Paper Award)
- [3] **Stable-TTS: Stable Speaker-Adaptive Text-to-Speech Synthesis via Prosody Prompting**
Wooseok Han*, Minki Kang*, Changhun Kim, Eunho Yang
IEEE International Conference on Acoustics, Speech, and Signal Processing (ICASSP), 2025

- [2] **CloudFixer: Test-Time Adaptation for 3D Point Clouds via Diffusion-Guided Geometric Transformation**
Hajin Shim*, **Changhun Kim***, Eunho Yang
European Conference on Computer Vision (ECCV), 2024
- [1] **SGEM: Test-Time Adaptation for Automatic Speech Recognition via Sequential-Level Generalized Entropy Minimization**
Changhun Kim, Joonhyung Park, Hajin Shim, Eunho Yang
Conference of the International Speech Communication Association (INTERSPEECH), 2023 (**Oral**)

Workshop Papers

- [3] **Structure-Aware Set Transformers: Temporal and Variable-Type Attention Biases for Asynchronous Clinical Time Series**
Joohyung Lee, Kwanhyung Lee, **Changhun Kim**, Eunho Yang
ICLR Workshop on Time Series in the Age of Large Models, 2026
- [2] **DeltaSHAP: Explaining Prediction Evolutions in Online Patient Monitoring with Shapley Values**
Changhun Kim*, Yechan Mun*, Sangchul Hahn, Eunho Yang
ICML Workshop on Actionable Interpretability, 2025
- [1] **AdapTable: Test-Time Adaptation for Tabular Data via Shift-Aware Uncertainty Calibrator and Label Distribution Handler**
Changhun Kim*, Taewon Kim*, Seungyeon Woo, June Yong Yang, Eunho Yang
NeurIPS Workshop on Table Representation Learning, 2024

Awards and Honors

Awards

Best Paper Award, CIKM 2025 Workshop on Human-Centric AI	Nov. 2025
Top Reviewer (206/10,943=1.9%), ICML 2025	May 2025
Silver Prize, Korean Undergraduate Mathematics Competition	Jan. 2022

Honors

Korean Government Scholarship Program for Study Overseas, National Institute for International Education (NIIED)	Fall 2026 – Present
Full Doctoral Assistantship, University of Wisconsin–Madison	Fall 2026 – Present
Dongwon Full Master’s Scholarship, Dongwon Group	Spring 2022 – Fall 2023
Magna Cum Laude, KAIST College of Engineering	Feb. 2022
National Full Undergraduate Scholarship, Korea Student Aid Foundation (KOSAF)	Spring 2017 – Fall 2021
KAIST Convergence AMP Scholarship, KAIST School of Computing	Mar. 2019

Teaching Experience

Teaching Assistant at UW–Madison

CS220 Data Science Programming I	Fall 2026
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Mentoring Experience

Hyeongwon Jang, Ph.D. Student at KAIST	Oct. 2024 – May 2026
Vishal Srivastava, Ph.D. Student at NUS	Aug. 2025 – Apr. 2026
Wooseok Han, Researcher at AITRICS	Mar. 2024 – Nov. 2024
Taewon Kim, Undergraduate Student at KAIST	Mar. 2023 – Nov. 2023
Sungwoo Cho, Undergraduate Student at KAIST	Sept. 2022 – Apr. 2023

Academic Service

Journal Reviewer: TMLR, TNNLS, Neurocomputing, IJCV

Conference Reviewer: NeurIPS, ICML, ICLR, AISTATS, AAAI, IJCAI, ACL, EMNLP, ICASSP, LoG

Invited Talks

Delta-XAI: A Unified Framework for Explaining Prediction Changes in Online Time Series Monitoring

IEIE Workshop on AI & Semiconductor Technologies for Future Healthcare

May 2026